AIS Graph Analytics

Architecture and operations

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Current live, historical and graph platform. Includes automatic PyFlink startup, durable checkpoint recovery, ClickHouse derived streams and the six-tab Metabase dashboard.

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This document is the operational and architectural companion to the repository root `README.md`. It reflects the deployed streaming, analytical and graph platform. The English and Russian PDFs are regenerated from the current English and Russian Markdown, using the shared architecture asset. See PDF regeneration below. Documentation reviewed: 2026-10-14.

1. Platform overview

AIS Graph Analytics combines two ingestion modes:

- **Live:** BarentsWatch AIS -> Python producer -> Kafka.
- **Historical lakehouse:** MinIO source -> PySpark -> Iceberg raw/curated -> Trino, with manual Airflow orchestration.
- **Retained legacy historical path:** HAIS GeoParquet -> Airflow -> ClickHouse.

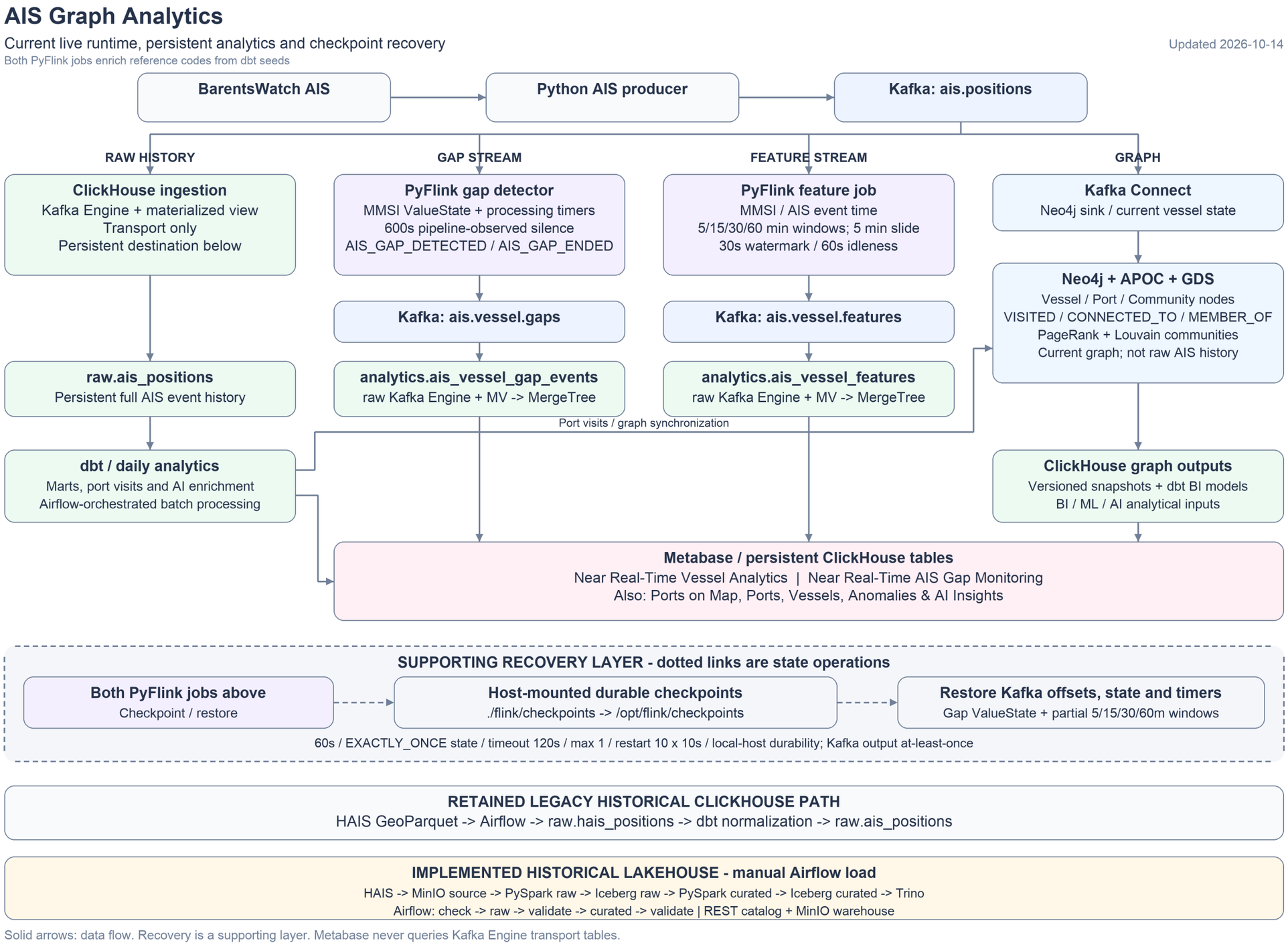
The live and legacy paths feed canonical ClickHouse history used by dbt; the lakehouse keeps its raw/curated tables in Iceberg. Neo4j stores graph/current-state structures, GDS produces graph metrics, OpenAI enriches selected daily vessel anomalies, and Metabase provides user-facing analytics.

The design separates responsibilities deliberately:

Concern	System of record / primary engine
Raw event history	ClickHouse
Historical lakehouse	MinIO + Iceberg REST + PySpark + Trino
Analytical transformations	dbt on ClickHouse
Current vessel graph state	Neo4j
Port/community graph analytics	Neo4j GDS + ClickHouse snapshots
Live transport	Kafka
Streaming windows and silence detection	PyFlink 2.2.1
Batch/daily orchestration	Airflow
AI interpretation	External OpenAI enrichment pipeline
Visualization	Metabase
CI/CD	GitHub Actions + GHCR

2. Architecture

Architecture: live and historical lakehouse paths



Solid arrows: data flow. Recovery is a supporting layer. Metabase never queries Kafka Engine transport tables.

Zoomable SVG · Editable diagram and regeneration. Solid arrows carry data; the separate dotted recovery layer carries state operations. The amber historical lakehouse path is implemented as a manual Airflow load.

2.1 Live path

```
BarentsWatch
  -> producers/ais
  -> Kafka topic ais.positions
     -> ClickHouse Kafka Engine / materialized view -> raw.ais_positions
     -> Kafka Connect Neo4j sink -> current Vessel nodes
     -> PyFlink gap detector -> ais.vessel.gaps
        -> raw Kafka Engine / MV -> analytics.ais_vessel_gap_events
     -> PyFlink multi-window features -> ais.vessel.features
        -> raw Kafka Engine / MV -> analytics.ais_vessel_features
```

The producer uses MMSI as the Kafka key and preserves source AIS fields. These are three distinct live paths: raw event history, stateful derived streams and graph analytics. ClickHouse owns historical records; Flink owns continuous stateful computation; the Neo4j sink merges current vessel state by MMSI. Graph relationships and GDS metrics complement the warehouse rather than replacing it.

2.1.1 Flink derived analytics and recovery

```
docker compose --profile streaming up -d
docker compose logs -f flink-job-submitter
```

The duplicate-safe one-shot submitter starts both AIS jobs after Kafka, JobManager, TaskManagers and task slots are ready. Both read one source topic,

`ais.positions`, keyed by MMSI, with their existing independent consumer groups.

- **Gap detector:** Flink ValueState and processing-time timers track the latest observation and active gap. After 600 seconds of pipeline-observed silence it emits `AIS_GAP_DETECTED`; the next observation emits `AIS_GAP_ENDED`, both to

`ais.vessel.gaps`. Processing time is intentional: detection must work without new AIS event timestamps. Source AIS `msgtime` is preserved descriptively in the payload and does not drive the silence timer.

- **Feature job:** AIS `msgtime` supplies event time. A 30-second bounded out-of-orderness watermark and 60-second input idleness govern keyed sliding 5/15/30/60-minute windows, with a five-minute slide. All branches emit to one topic, `ais.vessel.features`; `window_minutes` identifies the branch. Vessel name is the latest non-null name by event time, while navigation status and ship type come from the latest event-time observation, including null values.

The current production environment preserves:

```
FLINK_GAP_TIMEOUT_SECONDS=600
FLINK_FEATURE_WINDOWS_MINUTES=5,15,30,60
FLINK_FEATURE_SLIDE_MINUTES=5
FLINK_FEATURE_WATERMARK_SECONDS=30
FLINK_FEATURE_IDLE_SECONDS=60
FLINK_CHECKPOINT_INTERVAL_SECONDS=60
FLINK_CHECKPOINT_TIMEOUT_SECONDS=120
FLINK_CHECKPOINT_MIN_PAUSE_SECONDS=10
FLINK_CHECKPOINT_DIR=file:///opt/flink/checkpoints
FLINK_RESTART_ATTEMPTS=10
FLINK_RESTART_DELAY_SECONDS=10
```

Both jobs enrich reference codes from `dbt/seeds/ais_ship_types.csv` and

`dbt/seeds/ais_navigational_status.csv`. This shared reference data does not merge the two jobs' independent timing or state semantics.

ClickHouse persists the derived streams in `analytics.ais_vessel_features` and

`analytics.ais_vessel_gap_events`; Kafka transport tables and views remain in

`raw`. The first column, `activity_date`, uses UTC window-end dates for features and detection/end dates for gaps. Feature names use the latest non-null AIS name by event time within each window. Kafka Engine tables and materialized views stay in `raw` as ingestion infrastructure; the BI-facing destinations are persistent MergeTree tables. `raw.ais_positions` is the canonical source-history ReplacingMergeTree, with its replacement semantics documented

below.

`activity_date` is a ClickHouse storage/reporting convenience, not a new Flink payload field. Live Metabase cards use latest windows or rolling ranges rather than this historical date filter.

Both jobs checkpoint every 60 seconds to gitignored `./flink/checkpoints`, mounted at `/opt/flink/checkpoints` in all Flink services. Checkpoints use EXACTLY_ONCE state mode, a 120-second timeout, ten-second minimum pause, one concurrent checkpoint and retention on cancellation. Task failures use ten fixed-delay retries ten seconds apart. After a full cluster restart, the submitter restores missing jobs from retained checkpoints, including Kafka offsets, gap ValueState, processing-time timers and partial feature windows. A fresh start without a checkpoint instead begins at latest offsets. Without checkpoint restoration, keyed state, timers and partial windows are lost after a complete cluster restart. EXACTLY_ONCE applies to checkpointed state, not full end-to-end delivery. Kafka sinks use `AT_LEAST_ONCE`; ClickHouse ingestion also remains at-least-once, so replay can duplicate records. The local mount is not multi-host HA or protection against host/disk loss, and Kafka must retain the checkpoint's required offsets.

See the Flink operations and recovery tests and ClickHouse ingestion guide. The 2026-10-14 recovery validation passed a Flink-only restart and isolated state/timer/window tests; Kafka, ClickHouse, Neo4j, Airflow and Metabase stayed running.

Startup, reconciliation and manual restore

The submitter polls JobManager readiness, two registered TaskManagers, Kafka protocol connectivity and sufficient free slots. Each missing job needs one slot at the current parallelism of one. The shared registry and file lock record fixed Job IDs before submission, preventing duplicate submissions across retries and concurrent clients. Transitional jobs are tracked by ID; legacy jobs are adopted using their name plus source/operator plan. Only missing jobs are submitted with

`--pyFiles /opt/flink/jobs`; exit code 0 confirms both are RUNNING.

```
# Reconcile without restarting other services.
docker compose --profile streaming run --rm --no-deps flink-job-submitter
# Explicit Flink-only restart invokes the submitter via depends_on restart:true.
docker compose --profile streaming restart \
  flink-jobmanager flink-taskmanager flink-feature-taskmanager
```

The submitter is a one-shot startup client. After an abrupt Docker failure, rerun Compose startup/reconciliation; it is not a continuous supervisor. In-cluster failures use the Flink restart strategy. Complete cluster loss requires submission with a retained snapshot. The submitter selects finalized checkpoint metadata, uses `-s` with `-claimMode NO_CLAIM`, and fails on an invalid restore rather than silently discarding state. If no snapshot exists, it warns and starts fresh.

Manual restore is for an absent job only; replace the placeholders with a real completed checkpoint. Use the features directory and Python file for the other job.

```
docker compose exec -T flink-jobmanager /opt/flink/bin/flink run -d \
  -s file:///opt/flink/checkpoints/gap/<job-id>/chk-<id>/_metadata \
  -claimMode NO_CLAIM --pyFiles /opt/flink/jobs \
  -py /opt/flink/jobs/ais_gap_detector.py
```

2.2 Historical path

```
HAIS GeoParquet files
  -> Airflow historical ingestion
  -> raw.hais_positions
  -> dbt staging normalization
  -> canonical load macro
  -> raw.ais_positions
```

The HAIS landing layer is intentionally source-faithful. Normalization happens before loading into the canonical AIS table.

Implemented historical lakehouse

```
Verified host HAIS Parquet -> separate upload -> MinIO source
  -> PySpark -> iceberg.hais_raw.ais_positions
  -> PySpark curated transformation -> iceberg.hais_curated.ais_positions
  -> Trino validation and SQL
Airflow: source check -> raw -> validate -> curated -> validate -> summary
```

This path is deployed locally under the `lakehouse` profile. The preceding ClickHouse HAIS path is intentionally retained as legacy; neither it nor live Flink processing is part of the new DAG.

Component	Current role
MinIO	S3-compatible source bucket and separate Iceberg warehouse.
Iceberg REST catalog	Persistent SQLite catalog referencing Iceberg metadata in MinIO.
PySpark	One-shot local drivers for raw ingestion and curated transformation.
Trino	Single-node SQL engine with an 8 GiB heap and the shared Iceberg catalog.
Airflow	Manual-only `ais_hais_lakehouse_load` orchestration; no source upload.
Flink	Independent continuous live processing, not historical batch processing.

Sources live in `s3://ais-data/hais/source/`; table files/metadata live in

`s3://warehouse/`. Raw preserves all 14 source columns, including binary geometry. Byte/short fields widen to Iceberg integers; source GEOMETRY annotations require the compatible Parquet reader and explicit Arrow-derived schema. Curated columns:

Column	Type	Required
date_time	TIMESTAMP	Yes
mmsi	INT	Yes
longitude	DECIMAL(9,6)	Yes
latitude	DECIMAL(8,6)	Yes
navigational_status	INT	Yes
course_over_ground	DECIMAL(5,2)	No
true_heading	INT	No
speed_over_ground	DECIMAL(5,2)	No
rate_of_turn	INT	No
maneuver	INT	Yes
data_source	STRING	Yes
ais_class	STRING	Yes
message_type	INT	Yes
geometry	BINARY	Yes

Mappings: `date_time_utc` -> `date_time`, `status` -> `navigational_status`,

`maneuvre` -> `maneuver`, `msg_type` -> `message_type`. Course is retained only for `0 <= value < 360`; heading 511, speed 102.30 and turn -128 become NULL. There are no row filters, deduplication or enrichment. Coordinates round to six

decimals. Partitioning is only `day(date_time)`. Spark TIMESTAMP appears in Trino as `timestamp(6) with time zone`; STRING/BINARY appear as varchar/varbinary.

```
docker compose --profile batch --profile lakehouse build airflow spark
docker compose --profile batch --profile lakehouse up -d --wait --wait-timeout 300 airflow minio minio-init
   iceberg-rest trino
# Verify HAIS_UPLOAD_HOST_DIR before this separate upload:
docker compose --profile lakehouse run --rm hais-upload
docker compose exec -T airflow airflow dags unpause ais_hais_lakehouse_load
docker compose exec -T airflow airflow dags trigger --run-id hais_manual_1 ais_hais_lakehouse_load
python3 lakehouse/validate_curated.py
```

The DAG has `schedule=None`, `catchup=False`, one active run and serial tasks. Source checks retry twice; Spark loads once, with a two-minute delay. Validators have no retries; failures block downstream tasks. Spark runs in disposable containers through DockerOperator, not inside Airflow. Logs and graph/run state are available in the Airflow UI. Local ports: Trino 8090, REST 8181, MinIO S3 9002, console 9003; existing ports are unchanged.

Both full loads overwrite rather than append. Two manual runs on September 2026 validated 30 files and 23,295,797 rows in each layer. Required NULL rows and remaining sentinels are zero. Expected optional NULL counts: course 2,517,680, heading 4,511,883, speed 50,962 and turn 4,221,794. Validation also verifies schema, partition metadata and representative daily queries.

Limitations: local host/disk durability, SQLite REST fixture rather than a production catalog, no distributed HA/TLS, daemon control through Airflow's Docker socket, retained snapshots consuming disk, no cross-table transaction, no incremental loading and no simultaneous manual/DAG overwrites. Expected validation counts are September-specific. MinIO builds from pinned upstream source because community images were unavailable; Spark scripts must be rebuilt into the image for DockerOperator. See complete operating instructions.

2.3 Analytical / graph path

```
raw.ais_positions
  -> dbt vessel models
  -> daily features
  -> deterministic anomalies
  -> AI enrichment history
  -> daily enriched vessel view

raw.ais_positions
  -> port visit detection
  -> Neo4j port graph
  -> PageRank + Louvain
  -> ClickHouse graph snapshots
  -> Metabase
```

Kafka Connect supplies current Vessel nodes; the analytical port-visit workflow builds Vessel-to-Port `VISITED` and Port-to-Port `CONNECTED_TO` relationships. GDS PageRank and Louvain derive importance and communities, represented by Port-to-Community `MEMBER_OF` relationships. Exported graph snapshots in ClickHouse and dbt models are reusable BI/ML/AI analytical inputs. Metabase reads those persisted outputs, not Neo4j as a replacement raw-history store. See the graph pipeline.

3. Docker runtime

3.1 Compose profiles

Profile	Main service	Use
none	Kafka, Kafka Connect, ksqlDB, Kafbat UI, ClickHouse, Neo4j	Core runtime
`live`	AIS producer	Continuous BarentsWatch ingestion
`batch`	Airflow	Historical and daily analytics
`analytics`	Metabase	BI/dashboarding
`streaming`	Flink JobManager, two TaskManagers, checkpoint initializer and submitter	Stateful live features/gaps and recovery
`lakehouse`	MinIO, REST catalog, Trino, Spark and separate upload client	Historical raw/curated Iceberg loads

3.2 Project-owned images

The release workflow publishes three images to GHCR:

```
ghcr.io/konstantinlofichenko/ais-graph-analytics-airflow
ghcr.io/konstantinlofichenko/ais-graph-analytics-producer
ghcr.io/konstantinlofichenko/ais-graph-analytics-neo4j
```

The Flink image is built locally and is not currently included in GHCR releases.

The Neo4j image is customized because the project adds ClickHouse JDBC support on top of the upstream Neo4j image. Temporary `*-ci` image names are test artifacts and are not release packages.

3.3 Third-party images

Kafka, ClickHouse, Metabase, ksqlDB, Kafbat UI, and other upstream infrastructure are referenced directly from their official registries. They are intentionally not republished as project-owned packages.

4. Fresh installation

4.1 Repository and environment

```
git clone https://github.com/KonstantinLofichenko/ais-graph-analytics.git
cd ais-graph-analytics
cp .env.example .env
mkdir -p data/hais
```

Configure local secrets in `.env`. At minimum, set ClickHouse and Neo4j credentials. Configure BarentsWatch credentials only when live ingestion is required.

Never commit `.env`, connector passwords, or expanded configuration output that contains secrets.

4.2 Kafka Connect local config

```
cp neo4j/connectors/ais-sink.example.json neo4j/connectors/ais-sink.json
```

Use:

```
bolt://neo4j:7687
```

inside Docker. `localhost` from Kafka Connect refers to the Kafka Connect container itself and is therefore incorrect for Neo4j access.

4.3 Start core services

```
docker compose up -d --wait --wait-timeout 300
docker compose ps
```

4.4 Bootstrap

```
./scripts/bootstrap.sh
```

The bootstrap script:

1. Validates local connector configuration.
2. Creates/verifies Kafka topic `ais.positions`.
3. Creates/migrates the Flink-derived analytical tables without dropping legacy history, then applies additive ClickHouse migrations `003` through `010`.
4. Applies Neo4j constraints.
5. Creates/updates the Neo4j Kafka sink connector.

Migration `001_replacing_ais_positions.sql` is an existing-installation migration and is intentionally not part of fresh bootstrap.

5. ClickHouse data model

5.1 Canonical raw AIS

`raw.ais_positions` is the canonical durable event table. It is optimized for vessel/time access and uses event-time monthly partitions.

Important semantics:

- key analytical access: `(mmsi, msgtime)`;
- version column: `ingested_at`;
- engine: `ReplacingMergeTree(ingested_at)`;
- use `FINAL` for logical validation when replacement convergence matters;
- live and historical sources can overlap, so source coverage should be reviewed before large backfills.

5.2 Historical HAIS

`raw.hais_positions` preserves HAIS source values for auditability. dbt staging normalizes source details before canonical loading.

5.3 Derived stream history

Kafka topic	Persistent destination	ClickHouse consumer group
`ais.vessel.features`	`analytics.ais_vessel_features`	`clickhouse_ais_vessel_features`
`ais.vessel.gaps`	`analytics.ais_vessel_gap_events`	`clickhouse_ais_vessel_gaps`

Both destinations are MergeTree tables. Their Kafka Engine tables and materialized views remain in `raw`; BI reads persistent destinations. `activity_date` is the first column: features use `toDate(window_end)`; gaps use `toDate(gap_detected_at)` for AIS_GAP_DETECTED and `toDate(gap_ended_at)` for AIS_GAP_ENDED, all in UTC. DEFAULT expressions apply to old history without a full rewrite. Historical feature names absent from the original payload remain NULL.

The derived bootstrap script uses RENAME to preserve legacy table UUIDs/history when moving from `raw` to `analytics`. Conflicting old/new destinations fail before migration. Views are retargeted and reruns do not copy data again. ClickHouse

DDL uses literal topics and `ais-kafka:29092`; it does not dynamically read `.env`. Changing Flink topics alone therefore requires a matching ingestion DDL change. See the schema and monitoring guide.

5.4 Port visits and graph history

`analytics.port_visits` and `analytics.port_graph_metrics` use replacement/tombstone semantics. Stale logical records are marked with newer deleted versions rather than physically removed.

Downstream consumers must use the logical active state (`FINAL` plus the active/deleted filter where required).

6. dbt

6.1 Environment

```
python3.11 -m venv .venv-dbt
source .venv-dbt/bin/activate
python -m pip install -r dbt/requirements.txt
cp dbt/profiles.yml.example dbt/profiles.yml
export DBT_PROFILES_DIR="$PWD/dbt"
export CLICKHOUSE_USER=default
read -r -s -p 'ClickHouse password: ' DBT_ENV_SECRET_CLICKHOUSE_PASSWORD
printf '\n'
export DBT_ENV_SECRET_CLICKHOUSE_PASSWORD

dbt deps --project-dir dbt --profiles-dir dbt
dbt parse --project-dir dbt --profiles-dir dbt
dbt debug --project-dir dbt --profiles-dir dbt
```

Current pinned versions are dbt Core `1.12.5` and `dbt-clickhouse 1.10.3`.

6.2 Important models

Model	Role
`vessels`	Vessel identity/current analytical state
`vessel_daily_features`	Daily movement and quality features
`vessel_daily_anomalies`	Deterministic anomaly selection
`vessel_daily_enriched`	Daily features + valid current AI enrichment
`current_port_visits`	Logical current active port visits
`port_graph_metrics_enriched`	Graph metrics with reference dimensions
`port_graph_communities`	Community-level reporting table

Static AIS ship-type and navigation-status mappings are dbt seeds, not ad-hoc SQL embedded throughout models.

7. Retained legacy historical ClickHouse processing

Historical files are manually placed in `data/hais` with names such as:

```
hais_2026-09-01.snappy.parquet
```

The historical Airflow ingestion records audit/retry status. After successful source ingestion, dbt staging plus the canonical-load macro populates `raw.ais_positions`.

Date convention:

Operation	Start	End
HAIS file ingestion	Inclusive	Inclusive
Canonical dbt load	Inclusive	Exclusive UTC midnight
Daily analytics	Activity date / UTC window start	Next UTC boundary

Avoid uncontrolled overlapping re-loads: the canonical table can converge duplicate versions, but backfills can still replace attributes such as vessel names/types if the historical source is less complete than live data.

8. Daily Airflow orchestration

The main DAG is:

`daily_ais_pipeline`

Schedule:

`02:00 UTC daily`

An explicit `dag_run.conf.activity_date` can be used for deterministic reruns/backfills.

The daily orchestration coordinates the analytical chain rather than treating dbt, AI, graph, and dashboard data as independent manual jobs.

The actual task order is:

```
resolve_activity_date -> dbt_core -> dbt_vessel_daily_features
  -> ais_port_visits -> ais_gds_metrics -> ais_graph_metrics_export
  -> dbt_graph_models -> dbt_vessel_daily_anomalies -> ai_enrichment
  -> dbt_vessel_daily_enriched -> dbt_tests
```

The date is captured once from an explicit activity_date or yesterday UTC at master start. Child graph DAGs must be unpaused; their trigger tasks wait for success. Country-dependent graph enrichment remains optional. See Airflow operations.

Historical backfills should be sequential where graph/current-state updates are involved.

9. AI enrichment

9.1 Purpose

AI is used to summarize selected anomalous vessel-days, not to replace deterministic feature engineering or anomaly selection.

9.2 Cache and history

`analytics.vessel_ai_enrichment` retains response history including model, prompt version, exact input hash, token counts, response ID, and creation time.

The lookup requires an exact match of the relevant input identity, including date/model/prompt/input hash. This avoids presenting stale enrichment after source features change while preserving historical responses for auditability.

9.3 Bounded calls

`AI_ENRICHMENT_LIMIT` limits **new** API calls after cache hits are removed. CI does not make live OpenAI calls.

10. Neo4j and graph analytics

10.1 Current state vs history

Neo4j represents the current graph state. ClickHouse stores time-versioned graph snapshots for historical analysis.

10.2 Main relationships

```
(Vessel)-[:VISITED]->(Port)
(Port)-[:CONNECTED_TO]->(Port)
(Port)-[:MEMBER_OF]->(Community)
```

10.3 GDS

The graph workflow includes:

- PageRank for port importance;
- Louvain for community detection;
- deterministic community naming/labels based on representative ports;
- export of graph metrics to ClickHouse.

Community reporting is standardized around human-readable names while retaining the technical community ID for traceability.

11. Metabase

The main dashboard contains six tabs:

1. **Ports on Map** - port map and a recovered-gap vessel ranking.
2. **Ports** - centrality and community analytics.
3. **Vessels** - vessel activity, categories and last-known positions.
4. **Anomalies & AI Insights** - daily anomaly candidates and AI enrichment.
5. **Near Real-Time Vessel Analytics** - latest completed 5/15/30/60-minute Flink feature windows, active vessels, speeds, categories and navigation status.
6. **Near Real-Time AIS Gap Monitoring** - detected/recovered gaps over the last hour, and vessel rankings, durations and latest events over the last 24 hours.

The refreshed Metabase export and guide contains all 31 saved cards, layout, tabs, filters and mappings. The Window minute selector is mapped to feature cards 58-62; the features details table currently uses its own five-minute default. The new rolling/latest queries are independent of the historical Date and Last Seen filters. Historical cards use business

`activity_date` semantics where applicable.

12. CI

The CI workflow is intentionally isolated from external services and local developer state.

12.1 Jobs

Job	Validation
Shell and Compose	Shell syntax and all Compose profiles
Pipeline tests	Airflow image + isolated DAG/pipeline tests
Producer	Python producer tests + image build
dbt + ClickHouse	Fresh ClickHouse, init, migrations, CI fixture, seeds, selected model/test graph
ClickHouse + Kafka	Derived-topic ingestion, analytics ownership, names and idempotent migration
Kafka + Neo4j	Kafka topic/produce/consume + custom Neo4j startup/constraints/write/read

12.2 CI source fixture

The dbt integration test creates a deterministic `raw.countries` fixture only for CI because production `raw.countries` is normally owned by optional Airbyte ingestion. The fixture must remain CI-only and should not be added to normal production initialization.

12.3 External calls deliberately excluded

CI does not depend on:

- BarentsWatch;
- OpenAI;
- Airbyte control plane;
- a running local Mac environment;
- production Metabase.

This makes pull-request checks deterministic and reproducible.

13. Release workflow

A stable semantic tag matching `vMAJOR.MINOR.PATCH` triggers release automation.

Example:

```
git tag -a v1.0.3 -m "Release v1.0.3"
git push origin v1.0.3
```

The workflow:

1. Validates semantic version format.
2. Verifies the release commit belongs to `main`.
3. Runs the reusable CI workflow.
4. Builds the Airflow, producer, and custom Neo4j images.
5. Pushes versioned images to GHCR.
6. Creates a GitHub Release with generated notes.

Stable image tags include:

```
MAJOR.MINOR.PATCH
MAJOR.MINOR
MAJOR
latest
```

Do not move an already-published release tag. Use a new patch/minor/major version instead.

14. Published containers

```
docker pull ghcr.io/konstantinlofichenko/ais-graph-analytics-airflow:latest
docker pull ghcr.io/konstantinlofichenko/ais-graph-analytics-producer:latest
docker pull ghcr.io/konstantinlofichenko/ais-graph-analytics-neo4j:latest
```

Only project-owned images are published. Third-party images remain referenced from upstream registries.

15. Validation and operations

Useful project checks:

```
./scripts/status.sh
docker compose ps
docker compose logs --tail 100 airflow
docker compose logs --tail 100 ais-producer
docker compose logs --tail 100 neo4j
```

Kafka topic inspection and Flink readiness:

```
docker compose exec -T ais-kafka /opt/kafka/bin/kafka-topics.sh --bootstrap-server ais-kafka:29092 --list
curl -fsS http://localhost:8082/jobs/overview
docker compose logs --tail 100 flink-job-submitter
```

For each running job, inspect `/jobs/<job-id>/checkpoints` before a Flink-only restart. See the Flink recovery procedures.

Logical ClickHouse validation should prefer `FINAL` where ReplacingMergeTree replacement semantics matter.

16. Important data-quality semantics

- Ship type/name coverage depends on the source; source NULLs should not be silently presented as fabricated identities.
- AIS navigation status can be noisy; it is supporting evidence rather than a definitive motion label.
- Speed metrics exclude implausible values using the configured validity threshold.
- Stationary percentage is based on observations, not continuous elapsed-time interpolation.
- AI summaries describe the engineered inputs they receive; they do not establish behavior between AIS observations.
- Graph community IDs are technical identifiers; labels/names are user-facing descriptions.

17. Known limitation: midnight visit fragmentation

Port-visit detection processes bounded UTC windows. If one physical stay crosses midnight, it can appear as separate daily visit fragments because state is not persisted across daily windows.

Impact:

- possible visit overcounting;
- fragmented duration;
- small downstream differences in graph connections/metrics.

Production-grade alternatives are persistent detector state or continuous physical-visit detection followed by daily projection.

18. Troubleshooting

Container name conflict

```
docker ps -a
```

Remove only obsolete AIS test/manual containers. Do not blindly delete project volumes.

Kafka Connect cannot reach Neo4j

Use `bolt://neo4j:7687`, not `localhost`, inside Docker.

ksqlDB or Kafka Connect healthchecks

The Confluent images used by this project do not rely on `curl` for these healthchecks; the Compose configuration uses TCP checks where appropriate.

Neo4j custom image

The project-owned Neo4j image includes the ClickHouse JDBC JAR and is tested in CI with the plugin environment used by the normal Compose service.

dbt cannot connect

Confirm:

- dbt virtual environment is active;
- `DBT_PROFILES_DIR` points to the project dbt directory;
- `DBT_ENV_SECRET_CLICKHOUSE_PASSWORD` matches the local ClickHouse password;
- host/port are correct for the environment being tested.

19. Safe shutdown and cleanup

Stop services without deleting named volumes:

```
docker compose --profile '*' down
```

Do **not** add `-v` if data must be retained.

Temporary local CI images can be removed when no longer needed:

```
docker images --format '{{.Repository}}:{{.Tag}}' | grep -- '-ci'
```

Delete only disposable test images, not the project runtime images or upstream dependencies you still use.

20. Repository documentation map

Path	Purpose
`README.md`	Project overview and quick start
`docs/README.md`, `docs/README.ru.md`	English/Russian operational and architectural sources
`docs/assets/README.md`	Shared diagram and regeneration
`pipelines/hais/README.md`	HAIS ingestion and audit/recovery
`pipelines/historic_ais/README.md`	Historic REST repair/backfill
`airbyte/README.md`	Optional country enrichment exports
`data/ports/README.md`	Dated pilot port reference
`airflow/README.md`	Airflow-specific setup and orchestration
`dbt/README.md`	dbt configuration and transformations
`pipelines/port_visits/README.md`	Port-visit implementation
`pipelines/graph_metrics/README.md`	Graph metric export
`neo4j/gds/README.md`	GDS workflows
`flink/README.md`	Streaming jobs, durable checkpoints and recovery
`clickhouse/README.md`	Derived-topic ingestion and analytics tables
`metabase/README.md`	Six-tab dashboard and export/import workflow

21. Current status and roadmap

Current implementation includes:

- live + historical AIS ingestion;
- canonical ClickHouse history;
- dbt analytical models/tests;
- Airflow scheduled orchestration;
- Neo4j graph analytics;
- deterministic anomaly selection;
- AI enrichment with cache/history;
- durable Flink feature windows and gap detection;
- six-tab Metabase dashboarding;
- six-job CI;
- semantic GitHub Releases;
- GHCR publishing for Airflow, producer and custom Neo4j images. Flink is a local build.

The local MinIO/Iceberg/PySpark/Trino lakehouse and manual historical Airflow DAG are implemented. `dbt-trino`, incremental lakehouse ingestion and production distributed catalog/storage remain future work.

PDF regeneration

The English PDF is generated from this guide; the Russian PDF uses README.ru.md. Update both sources when platform behavior changes. Both embed the same architecture diagram. PDFs describe the reviewed implementation; runtime screenshots and dated validation results retain their explicit snapshot dates.

```
python3 -m venv /tmp/ais-docs-venv
/tmp/ais-docs-venv/bin/pip install reportlab pillow pymupdf
/tmp/ais-docs-venv/bin/python docs/render_pdfs.py
```

The renderer uses Arial on macOS or DejaVu fonts on Linux and writes the two existing PDF paths. Regenerate the architecture first if its source changed. Render every PDF page for visual review after changes; verify links, Cyrillic, tables, code wrapping and checkpoint/delivery limitations before sharing.